Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/14/2024
PI Name:	Lu, Xiao-Hong Ph.D.		
Project Title:	Develop a Novel Single-Cell Biodosimetry for Brain Genomic Instability and Neurodegeneration to Predict Clinical Health Outcomes in Human Spaceflight Crews		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	71103	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	11/15/2020	**End Date:**	06/01/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 06/01/2025 per F. Hernandez/ARC (Ed., 2/12/25) NOTE: End date changed to 11/30/2024 per NSSC information (Ed., 11/15/23) NOTE: End date is 11/30/2023 (incorrectly listed in NSSC as 11/14/2021) per F. Hernandez/ARC (Ed., 7/27/21)		
Key Personnel Changes/Previous PI:	Per the PI, the list of Co-Investigators has changed. During the past reporting period, the team has been working with the newly established bioinformatics center at Louisiana State University for the project, and Dr. Cvek was not involved. Dr. Harrison has moved to a new position and will no longer serve on this project. (Ed., 9/17/23)		
COI Name (Institution):	Chancellor, Jeffery Ph.D. (Louisiana State University and A&M College)		
Grant/Contract No.:	80NSSC21K0273		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As NASA plans future exploration missions to the Lunar and Martian surfaces, realistic ground-based analog studies and more predictive biodosimetry are needed to assess whether the space radiation poses a detrimental risk of brain genomic instability and neurodegeneration that leads to late-onset behavioral deterioration for spaceflight crews. Implementing a recently developed method of recreating the intravehicular (IVA) radiation environment expected on spaceflight vehicles and habitats and a novel genetic sensor, this proposal addresses Research Topic 3 – Animal Biology Studies in support of Human Space Exploration and Sub-topic AB1-A – Behavior and underlying neural function in Appendix D: Solicitation of Proposals for Flight and Ground Space Biology Research. We propose to determine how the space environment and sex affect brain genomic stability and consequent age-related brain structure and function changes. Our studies will support Human Space Exploration, by contributing the first biodosimetry for quantifying brain DNA instability and neurodegenerative changes to predict clinical health outcomes in human spaceflight crews and the utility of available ground-based analogs to realize basic mechanisms that can lead to the development of biologic counter-measures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our studies will support Human Space Exploration, by contributing the first biodosimetry for quantifying brain DNA instability and neurodegenerative changes to predict clinical health outcomes in human spaceflight crews and the utility of available ground-based analogs to realize basic mechanisms that can lead to the development of biologic counter-measures.
Task Progress:	1. To trace space radiation-elicited brain genotoxic stress in vivo, a genetic sensor was developed to provide sensitive spatiotemporal visualization of genomic instability. The design was based on the observations that DNA damage can increase permissivity to viral transduction, and host cell DNA damage response is an innate antiviral defense mechanism, and is thus inhibitory to viral life cycles. The design of the sensor exploits the recombinant adeno-associated virus (AAV) genome-processing mechanism and the instability of a hypermutable repeat sequence to detect neuronal genomic instability and visualize neurodegenerative pathology with membrane-tethered fluorescence protein in postmitotic differentiated neurons. 2. Genetic sensors of genotoxic stress were introduced into mice noninvasively and systemically. The mice were subsequently exposed to ground-based galactic cosmic radiation (GCR) analogs at sustained lunar operation (15 cGy), deep space habitat (30 cGy), and Mars-mission (75 cGy) relevant doses (50) in low (0.75 cGy/min) and high (3.2 cGy/min) dose rates, administered as either 5-ion simplified Galactic Cosmic Ray simulation (SimGCRsim) or 1000 MeV/n Fe passed through a moderator block to model intravehicular spallation. Another cohort of mice (6 months of age) were irradiated with 30 or 75 cGy moderated iron or 5-ion GCRsim. Sensorimotor, cognitive, social, and emotional behavior were monitored at two time points: 2-4 months and 10-14 months after irradiation. The oldest mice in the study lived up to 24 months, which is comparable to approximately 70 human years, making this the first long-term study to assess late-life central nervous system (CNS) risks across the entire lifespan of mice. Exposure to two methods of GCRsims induced robust genetic sensor labeling. 3. The brain regions with the highest labeling of cells with genomic instability include the cortex, striatum, thalamus, brain stem, and olfactory bulb, indicating that space radiation exposure can induce chronic brain-wide genotoxic stress. While marked neurodegeneration was not evident in the aged mice, sustained neural dysfunction in cognitive, emotional, and sensorimotor domains was noted 14 months after exposure. Alterations in emotional state and challenged movement were present at early time points and appeared to resolve over time. Other deficits – such as a prevalent alteration of spontaneous activity and sensorimotor function – persisted for more than a year, with a progressive decline in working memory. 4. The membrane tethering farnesylated/myristoylated derivatives of fluorescence proteins of the genetic sensor allows sensitive visualization and detection of the subtle neuropathology that cannot be detected by the traditional stereology-based method to count the loss of the cell number. A loss of dendritic stubby and thin spines of striatal medium spiny neurons was observed in the mice exposed to two methods of GCRsims. Single-cell pathology analysis further revealed sensor-labeled cells enriched with DNA damage, oxidative stress, senescence, and disruption of protein homeostasis, suggesting accelerated brain aging. 5. We found that GCRsim exposures disrupt brain transcriptomics in the mice 14 months after irradiation. Mice irradiated with 5-ion-GCRsim showed significantly more differentially expressed genes (DEG) in comparison to the group exposed to a modulated iron beam. 6. We are finalizing the first-ever brain-wide mapping of genetic sensor-labeled neurons with genotoxic stress after GCRsim exposures. The images of Artificially Recognized sensor-labeled neurons were registered to the Allen Brain Atlas. 7. We finished systemically characterizing and quantifying neuroinflammation in multiple brain regions at three time points after GCRsim exposures. Reactive microgliosis (indicated by increased diameter and surface area of Iba1-expressing microglia) was observed brain-wide in 24-month-old and 13-month-old mice, as well as many brain regions of 5-month-old mice – indicating a persistent and age-dependent neuroinflammation.
Bibliography Type:	Description: (Last Updated: 09/26/2024)
Abstracts for Journals and Proceedings	Knott E, Grames M, Keys K, Tian X, Chancellor J, Harrison L, Lu X-H. "Simulated GCR exposure causes persistent sensorimotor and cognitive changes elucidated by a novel genetic sensor." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023

Abstracts for Journals and Proceedings	Tian X, Lu D, Knott E, Chancellor J, Harrison L, Lu X-H. "Synthetic sensor-actuator circuit as a cellular robot for genetic dissection of space brain to understand the pathogenic role of exposure to space radiation and conferring radioprotection." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Awards	Reece ELK. "EPSCoR Research Awards Program 2024: Modeling and targeting genomic instability of prostate cancer to evaluate and manage risks to astronaut health during deep space travel." May-2024
Awards	Lofton K. "LaSPACE Graduate Student Research Assistance (GSRA) Program." May-2024